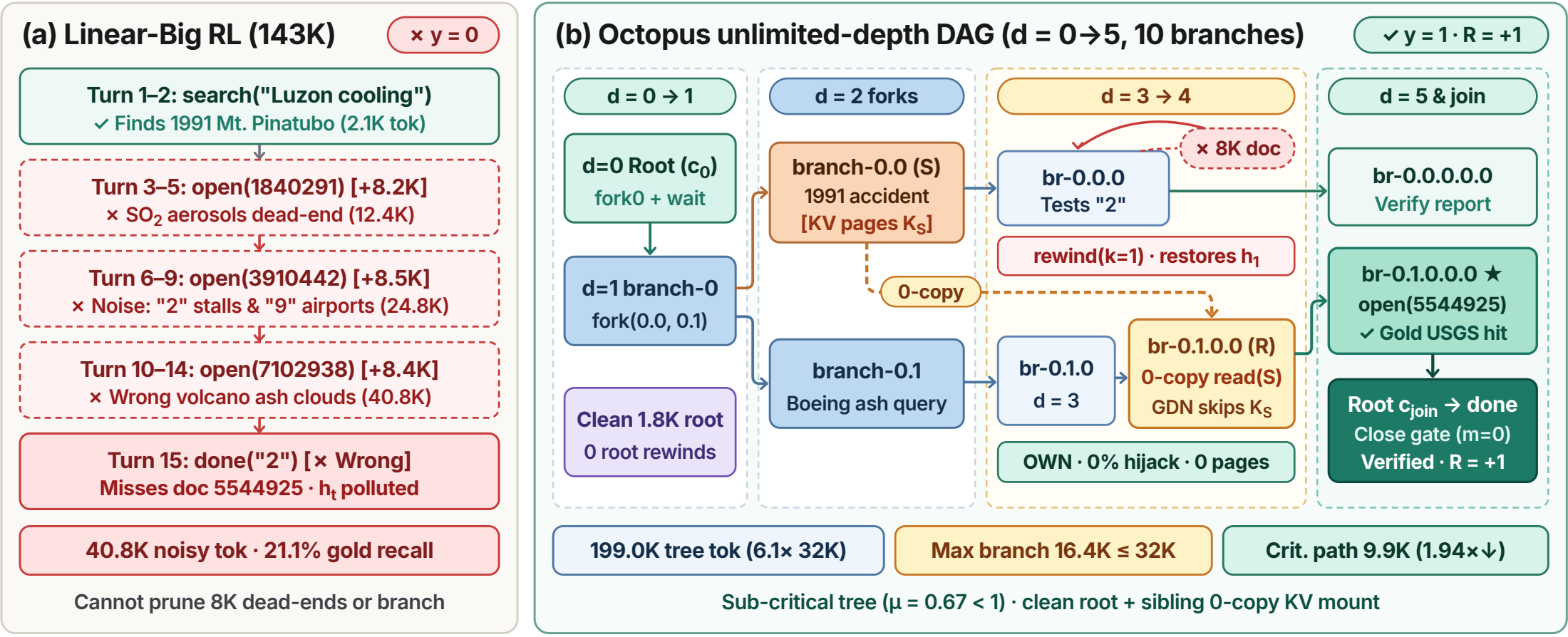
(a) Linear-Big RL (143K)
× y = 0
Turn 1–2: search("Luzon cooling")
✓ Finds 1991 Mt. Pinatubo (2.1K tok)
Turn 3–5: open(1840291) [+8.2K]
× SO_2 aerosols dead-end (12.4K)
Turn 6–9: open(3910442) [+8.5K]
× Noise: "2" stalls & "9" airports (24.8K)
Turn 10–14: open(7102938) [+8.4K]
× Wrong volcano ash clouds (40.8K)
Turn 15: done("2") [× Wrong]
Misses doc 5544925 · h_t polluted
40.8K noisy tok · 21.1% gold recall
Cannot prune 8K dead-ends or branch
(b) Octopus unlimited-depth DAG (d = 0→5, 10 branches)
✓ y = 1 · R = +1
d = 0 → 1
d=0 Root (c_0)
fork0 + wait
d=1 branch-0
fork(0.0, 0.1)
Clean 1.8K root
0 root rewinds
d = 2 forks
branch-0.0 (S)
1991 accident
[KV pages K_S]
branch-0.1
Boeing ash query
d = 3 → 4
× 8K doc
br-0.0.0
Tests "2"
rewind(k=1) · restores h_1
0-copy
br-0.1.0
d = 3
br-0.1.0.0 (R)
0-copy read(S)
GDN skips K_S
OWN · 0% hijack · 0 pages
d = 5 & join
br-0.0.0.0.0
Verify report
br-0.1.0.0.0 ★
open(5544925)
✓ Gold USGS hit
Root c_{join} → done
Close gate (m=0)
Verified · R = +1
199.0K tree tok (6.1× 32K)
Max branch 16.4K ≤ 32K
Crit. path 9.9K (1.94×↓)
Sub-critical tree (μ = 0.67 < 1) · clean root + sibling 0-copy KV mount